Industrial Safety Processes

Ropes

Slings

Chains

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Types of Fibre Rope

- Manila fibre
- Sisal fibre
- Synthetic
- Nylon
- Polyester
- Polyolefin
- Other natural fibres

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Fiber Rope Use

- Select type of rope for intended use and loads
  - Consider breaking strength and stretch
  - Note possible damage from load, such as sharp corners
- Environment is a major issue
  - Wet/dry
  - Chemical exposure
  - Heat/cold exposure

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Fiber Rope Use

- Natural fiber ropes loaded more than 50% of their rated strength will be permanently damaged
- Synthetic fiber ropes are likely to be damaged at 65% or more of their rated strength
- Splice ropes rather than tie them together
  - A good splice can retain up to 100% of rated strength
  - A knot retains only about 50% of the rated strength

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Fiber Rope Inspection

- Inspections are critical to safety
- Look for kinks, melts, cuts
- Use the “fingernail test”
  - Scratching weak fibers will cause them to come apart easily
  - Quick way of checking for chemical damage not otherwise readily visible

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Wire Rope

- Stronger, more versatile than fiber rope
- Many different types and classifications – need to choose the correct wire rope for the job
- Usually trade-offs in the selection of types of wire rope
  - More wires per strand = more flexible
  - Less wires per strand = more abrasion resistant

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http://image.made-in-china.com/2f0j00CBLTpSrdAPkM/Steel-Wire-Rope-6x36-FC-.jpg

Wire Rope Construction

- Wire strands wrapped around core
  - Fiber core
  - Independent Wire Rope Core (IWRC)
  - Strand core

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Wire Direction and “Lay”

- Strands in rope may be laid to the right or left
- Wires in the strands may be laid in the same direction as the strand or not
  - Opposite direction = regular lay
  - Same direction = lang lay

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Wire Rope Issues

- May be overloaded due to overconfidence in the lines
- Damage may be difficult to spot
- Examination of rope is critical – detailed measurements often required
- Trained personnel should inspect rope

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Ropes on Sheaves and Drums

- Important to make sure right rope is used for the drums and sheaves in a lifting system
- Too narrow sheave can cause rope fatigue
- Improper laying on of the rope may cause crushing and distortion
- Ensure that required wraps remain on the drum at all times (usually 5, but can vary)

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Wire Rope Fittings

- Attachments and connectors vary based on use
- Should be installed by manufacturer
- Should not be repaired by anybody other than manufacturer
- Use and application will vary by type of rope, construction, and manufacturer
- Clips – “don’t saddle a dead horse”

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Chains

- The strength and use of chains for lifting depends on the material that it is made of, how the chains are attached, and how they are inspected and maintained.
- Can be very good for harsh environments and conditions
- Need to be properly inspected – link by link

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Synthetic & Metal Mesh Slings

- Need to be inspected regularly
- Potential for abrasion and wear that could lead to catastrophic sling failure
- Synthetic mesh usually used to protect load from damage
- Metal mesh usually used to protect load that may at the same time damage the sling

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Main Issues Using Slings

- Must not be overloaded
- Angle of the lift has a major impact on lifting capacity of the rope/sling/chain
- Use of slings should conform to the type of load being lifted and the potential for losing the load if there is a shift
- Should be inspected before each lift to ensure safety

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